Test 2 Guidelines

Test Date: April 30.
Test materials: Chapter 3 ($\mathcal{S}23 - \mathcal{S}27$) and partial $\mathcal{S}31, 32$ from Chapter 4.
It is close-book and calculator is allowed.
Summary sheets (two pages, both sides ok) of theorems/formulas are allowed.

Please review the concepts and theorems in the above sections and homework sets 7-11.

Focus on the following concepts and questions:

1. Is a topological space connected? Or does a space have a separation?
2. Is a space path-connected?
3. Is a given space compact?
4. Is a given subset closed? Compact?
5. Is a given ordered set a linear continuum?
6. Given a pair of points in a space, find a path between them.
7. Find an open covering of a subset of a space (finite or infinite).
8. Given a compact space covered by collection of open sets, obtain a finite open covering from the collection.
9. Discuss compactness and closeness of a subspace by applying related theorems.
10. Is a given map a closed map?
11. Topological properties carried over by continuous functions.
12. Uniformly continuity via Lebesgue number.
13. In a metric space, find distance between two points, or between one point and a set.
14. Concepts of ϵ-neighborhood of a subset and ϵ-ball in a metric space.
15. Concepts of limit point and isolated point.
16. Discuss regular, normal, and Hausdorff properties and their relations (separation axioms).

Important: Review Homework sets #7-11 and the proofs of theorems in the textbook (not the very long ones)

Office Hours (in office):
Wednesday, April 29: 2:00-4:00 pm.
Thursday, April 30: 10:00- 11:30; 4:10-5:10 pm (in office).